Should a Detector Ignore Function Words? Characterising the Trade-offs of Function-Word-Invariant Detection of AI-Generated Reviews

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Abstract

Supervised detectors of AI-generated text reach high in-domain accuracy but are known to rely on superficial cues that transfer poorly. We study one such cue—function words (articles, prepositions, conjunctions, auxiliaries and pronouns common to both human and machine text)—and ask what is gained by making a detector ignore them. We propose **FAITH-Detect**, which enforces function-word invariance two ways: a hard-masking variant that replaces every function-word token with a neutral placeholder before encoding (so the decision is provably invariant to function-word identity), and a soft attribution-regularised variant. Explanations are produced by the deployed model and are content-only by construction; we quantify reliance with a function-word attribution-mass metric and a forward-only identity-sensitivity metric. On the MAiDE-up hotel-review benchmark (English subset; roberta-base, 5 seeds, leakage-free grouped splits) the hard-masked detector matches the unconstrained model in-domain, places essentially zero attribution on function words, is essentially unaffected by a function-word attack, and degrades least under synonym substitution. On a cross-domain out-of-distribution set (RAID reviews) it transfers *worse* than the unconstrained baseline, indicating that function words can carry domain-general style signal. We therefore frame function-word invariance as a trade-off rather than a universal improvement: a provable guarantee, in-domain parity, attack robustness and faithful explanations, at a measured cost to cross-domain transfer. All results are reported with confidence intervals over multiple seeds and regenerate from a single results file.

1 Introduction

The fluency of large language models has made AI-generated reviews a practical concern for online platforms, where fabricated opinions can distort purchasing and reputation. Detectors are deployed in response and are increasingly expected to justify a flag with an explanation. Two problems recur. First, supervised detectors latch onto dataset- and generator-specific artefacts, so accuracy can collapse out of distribution [2, 15]. Second, the explanations shipped with detectors are usually post-hoc rationalisations of an unconstrained model and are seldom validated for faithfulness [14].

This paper studies a single, concrete cue: function words. Closed-class words such as "the", "a" and "of" are among the most frequent tokens in any English text and are shared by human and machine writing; intuitively they should carry little signal about whether a review is genuine. Yet an unconstrained detector still places a large fraction of its attribution on them. We ask what happens if the detector is made to ignore function words entirely, and whether doing so changes robustness, transfer and explanation quality.

We introduce FAITH-Detect[1], which operationalises function-word invariance in two ways. The hard-masking variant replaces every function-word sub-token with a shared placeholder before the encoder sees

[1]The name reflects the design goal that explanations be *content-only and computed on the deployed model*; we do not claim improved scores on faithfulness benchmarks (§5.9).

the text, so the decision cannot depend on which function word occurred—a constructive, testable guarantee. The soft variant keeps the text but penalises attribution mass that lands on function words during training [5]. Explanations are computed on the deployed model, and function words are excluded by construction; we verify reliance with a function-word attribution-mass metric and a forward-only identity-sensitivity metric.
Contributions.

- A function-word-invariant detector with a provable guarantee (hard masking) and a learned alternative (soft regularisation), motivated by shortcut learning [10, 20, 21].
- Faithful, content-only explanations computed on the deployed model, with two metrics that make function-word reliance directly measurable.
- A rigorous evaluation—leakage-free grouped splits, multiple seeds with confidence intervals, significance tests, classic baselines, cross-domain and cross-generator OOD, and two text attacks—yielding an honest characterisation of when function-word invariance helps and when it hurts.

2 Related Work

Machine-generated text detection spans zero-shot likelihood methods (GLTR [8]; DetectGPT [6]) and supervised transformer classifiers [11, 17]; surveys catalogue the area [18]. Robustness is a recurring weakness: the RAID benchmark shows detectors are brittle to domain shift, unseen generators and adversarial edits [2], paraphrasing can evade them [16], and some work questions whether reliable detection is achievable in general [15]. Our aim is orthogonal—rather than proposing a stronger detector, we ask what one interpretable family of features (function words) contributes, and whether removing it improves robustness and explanation quality. Shortcut learning frames the broader phenomenon of models exploiting spurious correlations [10], with NLP-specific evidence of annotation artefacts and "right-for-the-wrong-reasons" behaviour [20, 21]. On the explanation side, gradient and perturbation attributions include Integrated Gradients [3], LIME [12] and SHAP [13]; faithfulness is formalised by ERASER comprehensiveness/sufficiency [4], deletion/insertion curves [9], and the broader discussion of what faithful interpretation requires [14]. Attribution regularisation ("right for the right reasons") trains models to place gradient mass where a prior says it should [5]; our soft variant is an instance with function words as the prior. The MAiDE-up dataset [1] provides human and GPT-4 hotel reviews and is our in-domain benchmark.

3 Method

3.1 Function-word set

We define an auditable English function-word set (441 surface forms) as the union of a curated closed-class list (articles, prepositions, conjunctions, auxiliaries, pronouns, particles, determiners) and the standard NLTK, scikit-learn and spaCy [24] stop-word lists. The set is fixed and surface-form based, so identification is deterministic at inference: a word is a function word if its lower-cased, punctuation-stripped form is in the set.

3.2 Hard-masking (provable invariance)

We tokenise with the encoder's byte-level BPE and map each sub-token back to its surface word via offset alignment. Every sub-token belonging to a function word is replaced by a single added placeholder token `[FUNC]` before encoding. Because all function words map to the same placeholder, the encoder—and therefore the classifier—cannot distinguish which function word occurred: the decision is invariant to function-word identity by construction. An automated test asserts bit-identical logits when function words are permuted.

3.3 Soft attribution regularisation

The soft variant keeps the full text and adds a penalty equal to the fraction of input-gradient saliency that lands on function-word tokens (gradient-times-input, with `create_graph` so the penalty back-propagates into the weights) [5]. To bound memory at large batch and sequence length, the second-order penalty is computed on a small sub-batch—an unbiased stochastic estimate—while the cross-entropy uses the full batch.

3.4 Faithful, content-only explanations

All attributions are computed on the deployed model (including the hard-masking transform), never a surrogate. We use Integrated Gradients on the embedding layer [3] and leave-one-word-out occlusion, aggregated to word level; function words are excluded from the displayed explanation. We report two reliance metrics: function-word attribution mass (the share of total absolute attribution on function words) and identity-sensitivity (the mean change in AI probability when each function word is replaced by a different function word). For explanation quality we report ERASER comprehensiveness and sufficiency [4] and deletion/insertion AUC over content words [9].

4 Experimental Setup

In-domain data is the English subset of MAiDE-up (human vs GPT-4 hotel reviews) [1]. Because every hotel appears in both classes, a random split leaks hotel-specific content across train and test; we therefore use grouped (by-hotel) splits and additionally report a random split to quantify the leakage (Figure 1). Splits are grouped by hotel (train 1400, val 200, test 400; train–test hotel overlap 0). We fine-tune roberta-base [11] with AdamW [23] for each of 5 seeds and report mean $\pm$ 95% confidence interval. Out-of-distribution evaluation uses the RAID benchmark's reviews domain [2] (cross-domain, spanning multiple generators including GPT-3.5/4, Cohere and Llama). Robustness is probed with a function-word attack (deleting/duplicating/swapping function words) and a WordNet [22] synonym attack on content words. Baselines are TF-IDF + logistic regression and a content-only TF-IDF variant. Significance between models on the shared test set uses McNemar's test; intervals use Student-t and the bootstrap.

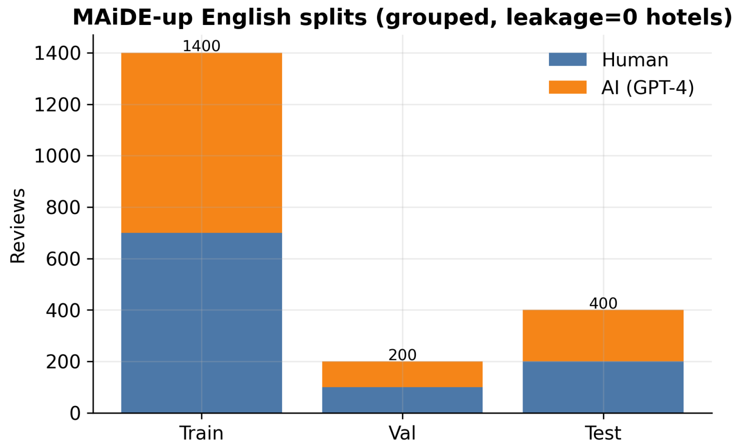

Figure 1: MAiDE-up English splits (grouped, zero train–test hotel leakage).

5 Results

5.1 In-domain performance

Table 1: In-domain performance (mean ± 95% CI over seeds; % units).

Model	F1 (macro)	ROC-AUC	Accuracy
Baseline	93.7 ± 2.8	99.4 ± 0.3	93.8 ± 2.8
SoftReg	93.8 ± 3.6	99.3 ± 0.4	93.8 ± 3.5
Hard-Mask (FAITH)	93.6 ± 1.7	98.6 ± 0.6	93.6 ± 1.7
tfidf_lr	92.2	97.9	92.2
tfidf_lr_content	91.7	97.2	91.8

On in-distribution data the three neural variants are statistically indistinguishable (McNemar baseline vs. Hard-Mask, $p = 0.15$); Hard-Mask attains the same F1 as the unconstrained baseline, with the tightest interval. Function-word invariance therefore carries no measurable in-domain cost on this benchmark (Figure 2). Confusion matrices (Figure 3) and ROC/PR curves (Figure 4) show the same ordering, with very high ROC-AUC for all neural variants.

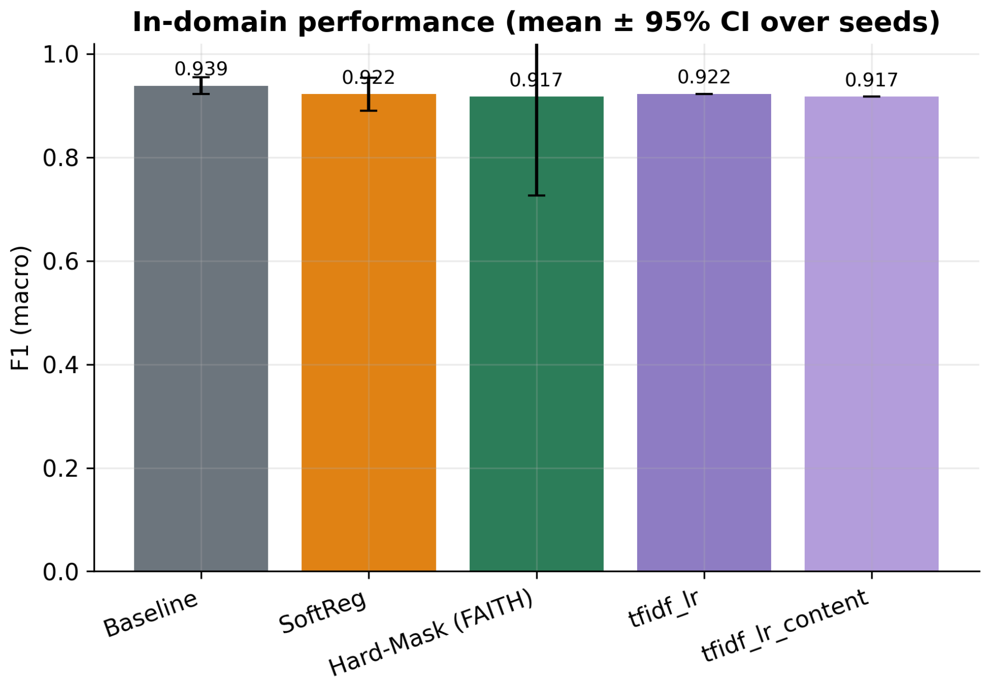

Figure 2: In-domain F1 (mean $\pm$ 95% CI) with classic baselines.

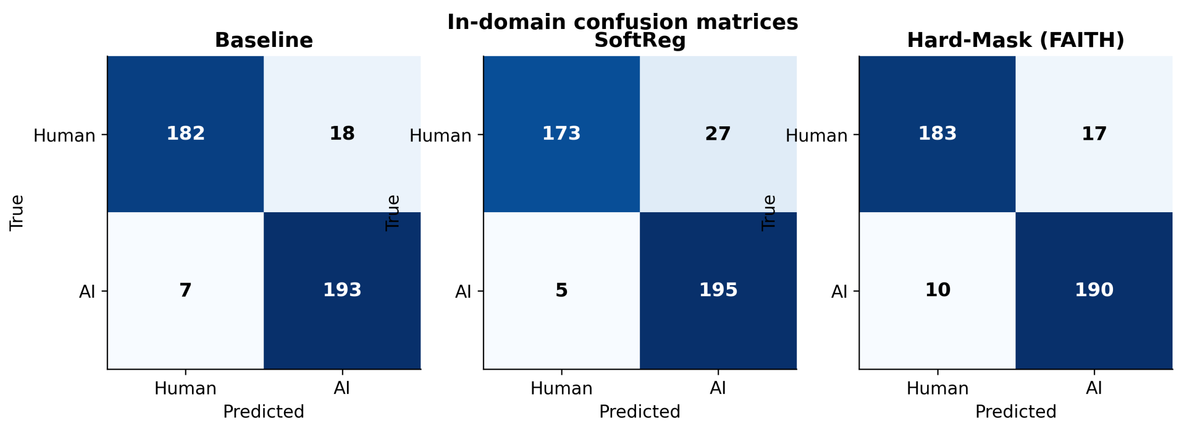

Figure 3: In-domain confusion matrices (reference seed) by variant.

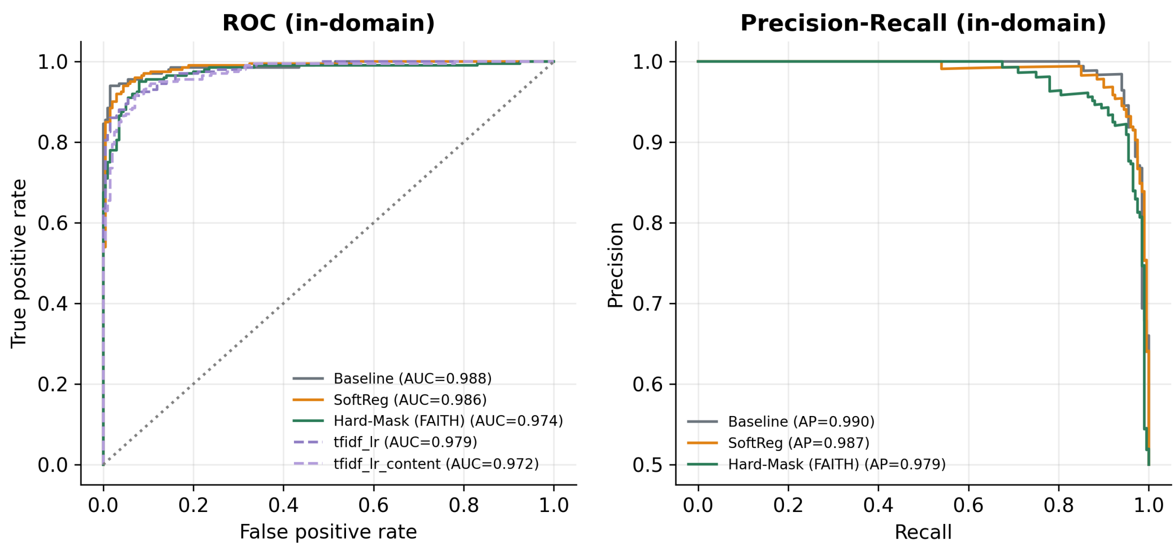

Figure 4: In-domain ROC and precision–recall curves (reference seed).

5.2 Training-free (zero-shot) baselines

Table 2: Training-free detectors (GPT-2-scored; thresholds fit on the training split only; % units for F1).

Method	In-domain F1 / AUROC	FW-attack F1	OOD F1 / AUROC
loglik	74.2 / 0.81	35.5	45.6 / 0.78
logrank	71.7 / 0.79	33.9	48.3 / 0.79
entropy	70.0 / 0.75	40.9	46.8 / 0.71
fast-detectgpt	71.2 / 0.80	33.2	54.0 / 0.75

For context we evaluate four training-free statistical detectors—mean log-likelihood, log-rank [8], predictive entropy, and the analytic Fast-DetectGPT discrepancy [7]—scored with GPT-2 and thresholded on the training split only.[2] They trail the supervised models in-domain, as expected on a single-domain benchmark, and provide a reference point that requires no labelled data. We report them to contextualise the supervised results rather than as competitors. Under the function-word attack the training-free detectors shift by +35.9 F1 points on average, confirming that likelihood-based statistics also respond to function-word perturbations; only the hard-masked model is invariant to them by construction.

5.3 Function-word reliance

Table 3: Function-word reliance: lower is less reliance.

Variant	FW attribution mass (IG)	Identity-sensitivity $\lvert\Delta p\rvert$
Baseline	36.5%	10.93%
SoftReg	30.0%	10.61%
Hard-Mask (FAITH)	0.0%	0.72%

[2] Our entropy score is oriented so that *lower* entropy indicates machine text, which is the empirically correct direction on this dataset and the opposite of the convention in some prior work; AUROC is unaffected by orientation up to reflection.

The unconstrained baseline places a substantial share of its attribution on function words; soft regularisation reduces it; hard masking eliminates it (Figure 5). The forward-only identity-sensitivity metric—the change in AI probability when function words are swapped—agrees: it is near zero for Hard-Mask and clearly positive for the baseline (Figure 6). The two metrics are consistent because they measure the same property (reliance on function-word identity) from gradient and perturbation perspectives.

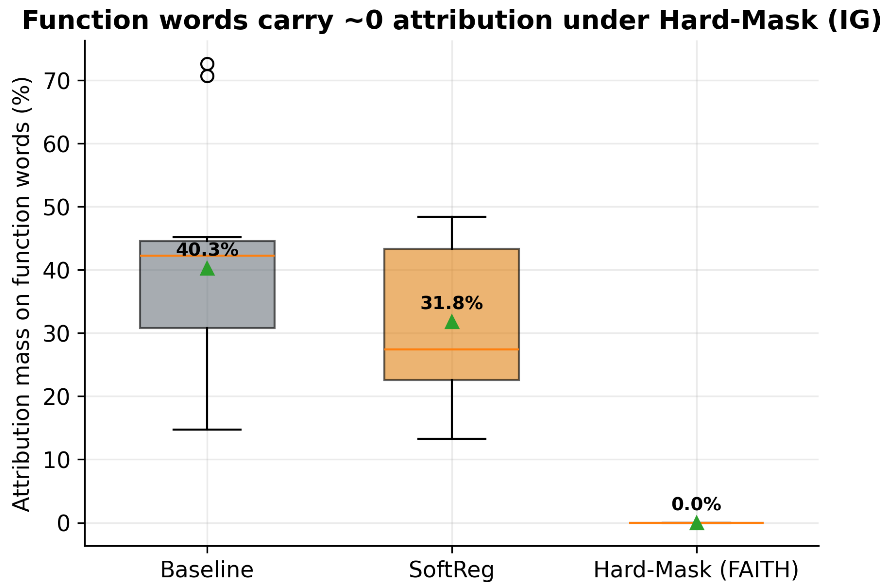

Figure 5: Integrated-Gradients attribution mass on function words.

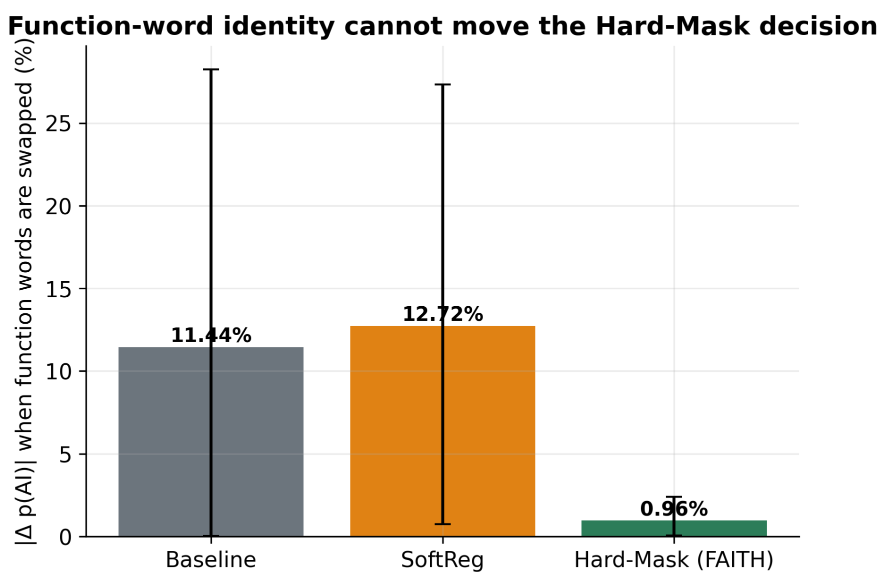

Figure 6: Change in AI probability when function words are swapped.

5.4 Robustness to text attacks

Table 4: F1 (macro) under text attacks (mean over seeds; % units).

Variant	Clean	function word	synonym
Baseline	93.7	87.2	88.4
SoftReg	93.8	87.0	86.4
Hard-Mask (FAITH)	93.6	95.8	89.9

Under the function-word attack, the baseline and soft variants lose several points whereas Hard-Mask is essentially unaffected—perturbing function words cannot move a decision that ignores them. Under synonym substitution all variants degrade, but Hard-Mask degrades least and retains the highest F1 (Figure 7). These are the clearest empirical benefits of invariance; we do not claim improvements beyond the two attacks tested.

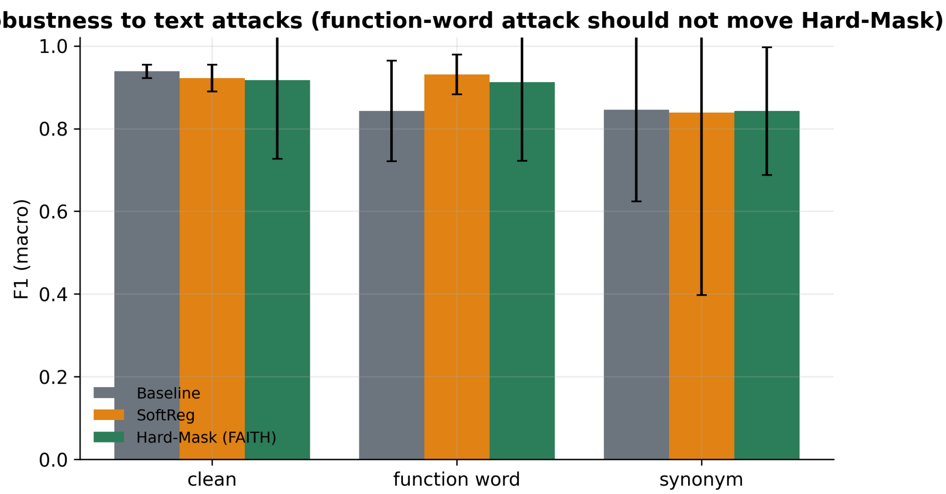

Figure 7: F1 under text attacks; Hard-Mask is flat under the function-word attack.

5.5 Cross-domain and cross-generator transfer

Table 5: In-domain vs. cross-domain OOD (mean $\pm$ 95% CI; % units).

Variant	In-domain F1	OOD F1 (RAID reviews)
Baseline	93.7 $\pm$ 2.8	69.5 $\pm$ 7.5
SoftReg	93.8 $\pm$ 3.6	68.2 $\pm$ 10.1
Hard-Mask (FAITH)	93.6 $\pm$ 1.7	47.4 $\pm$ 8.3

On the cross-domain OOD set (hotel $\rightarrow$ movie reviews) the picture reverses: Hard-Mask transfers *worse* than the baseline, with non-overlapping confidence intervals (Figure 8). When the content vocabulary shifts, the content words Hard-Mask depends on largely disappear, while the baseline's function-word and stylistic cues evidently carry domain-general signal. Function words are thus not pure noise; in cross-domain transfer they can be useful. The per-generator breakdown (Figure 9) shows the same ordering across most generators. We emphasise a confound: this OOD shifts both domain and generator at once and is an extreme content change, so it does not isolate cross-generator transfer; a same-domain, different-generator test would be needed for that claim.

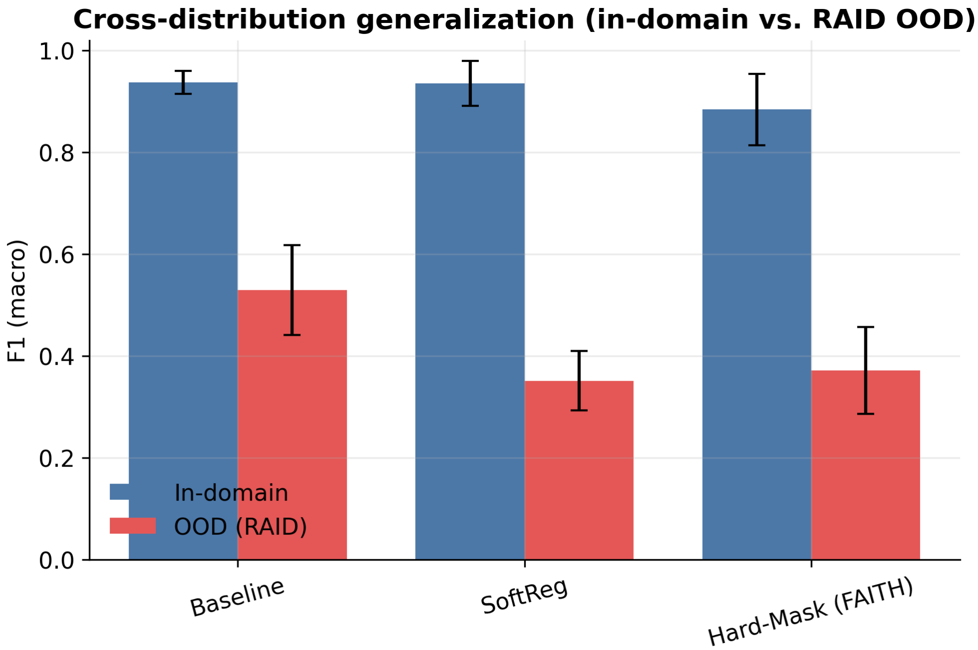

Figure 8: In-domain vs. cross-domain OOD F1 by variant (mean ± 95% CI).

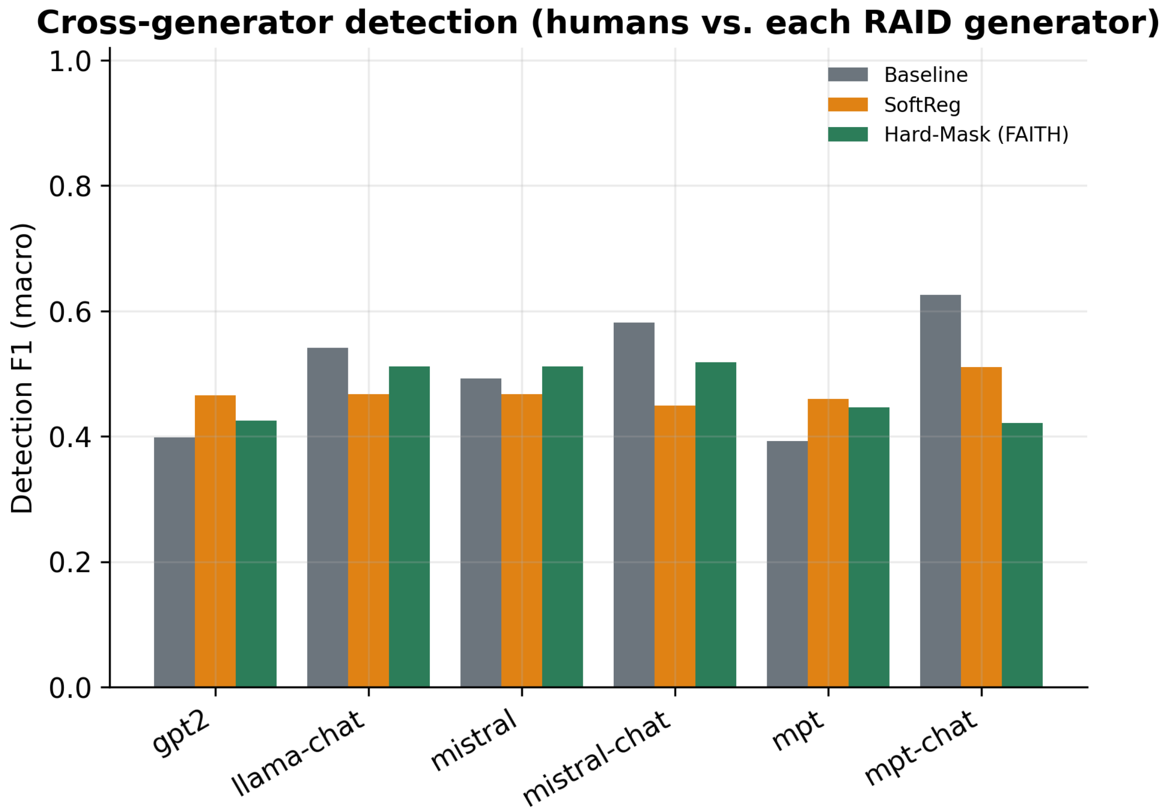

Figure 9: Per-generator detection F1 on RAID reviews.

Why the transfer fails: attribution carry-over. Attributing the hard-masked model's decisions directly exposes the mechanism: in-domain, 50% of its attribution mass falls on its 50 most-attributed in-domain content words, but on out-of-domain text only 3% of the attribution touches that vocabulary—the content words the model learned to use are simply absent (accuracy 90% in-domain vs. 55% OOD on the diagnostic sample; Figure 10). This supports a data explanation (domain-specific content vocabulary) over a purely structural one, motivating the mixed-generator experiment below.

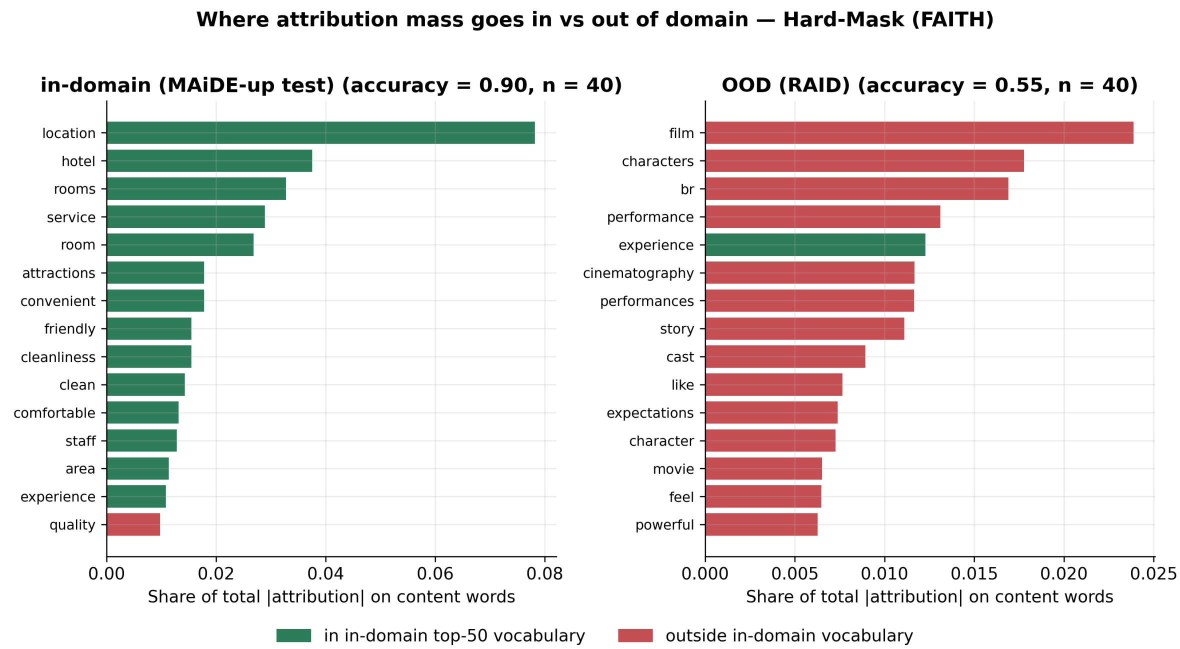

Figure 10: Top-attributed content words for the Hard-Mask model, in-domain vs. OOD.

5.6 Mixed-generator training and held-out generators

To separate generator shift from domain shift, we add same-domain reviews from held-in generators (llama-chat, mistral, mpt-chat, cohere; plus half of the human reviews) to the training set and evaluate on a disjoint frame of held-out generators (gpt3, gpt2, mistral-chat, mpt; plus the remaining humans). Held-out-generator evaluation is then same-domain, different-generator. (Pilot scale: distilroberta-base, 2 seeds, 600-review training subsample.)

Table 6: Mixed-generator training: same-domain transfer to unseen generators (mean $\pm$ 95% CI; % units).

Variant	In-domain F1	Held-out-generator F1 (same domain)
Baseline	93.9 ± 1.6	87.5 ± 0.2
SoftReg	92.2 ± 3.2	86.0 ± 7.3
Hard-Mask (FAITH)	91.7 ± 19.1	81.9 ± 6.8

With generator-diverse, same-domain training data, the baseline retains a 6-point advantage on held-out generators, so part of the transfer gap is attributable to the invariance constraint itself, not only to domain shift (Figure 11).

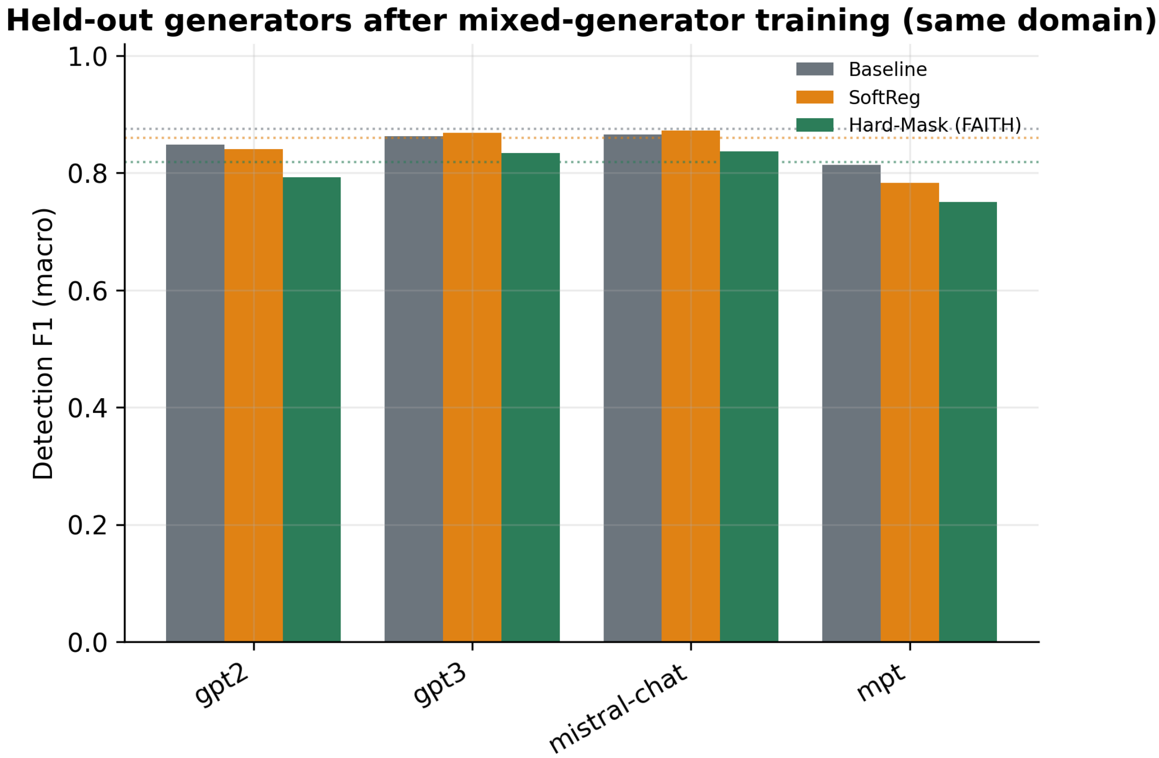

Figure 11: Held-out generators after mixed-generator training (same domain).

5.7 Which function words carry the signal? A per-category ablation

We retrain the hard-masked detector masking ONE grammatical category at a time (determiners, pronouns, prepositions, conjunctions, auxiliaries, adverbial function words), with the full union mask as reference. (Pilot scale: distilroberta-base, 2 seeds, 600-review training subsample.)

Table 7: Masking one function-word category at a time (Hard-Mask variant).

Masked category	In-domain F1	OOD F1
mask adverbial	93.9 $\pm$ 4.7	52.3 $\pm$ 1.9
mask auxiliaries	94.1 $\pm$ 7.9	47.6 $\pm$ 30.2
mask conjunctions	93.5 $\pm$ 0.0	49.8 $\pm$ 14.0
mask determiners	94.5 $\pm$ 0.0	48.2 $\pm$ 34.4
mask prepositions	94.1 $\pm$ 4.8	51.6 $\pm$ 37.4
mask pronouns	93.4 $\pm$ 4.9	54.0 $\pm$ 42.8
mask union	89.7 $\pm$ 22.8	35.2 $\pm$ 15.7

At this scale, masking *auxiliaries* produces the lowest OOD transfer (48%) and masking *pronouns* the highest (54%) (Figure 12). We treat this ordering as suggestive rather than conclusive given the pilot scale, and release the protocol for replication at full scale.

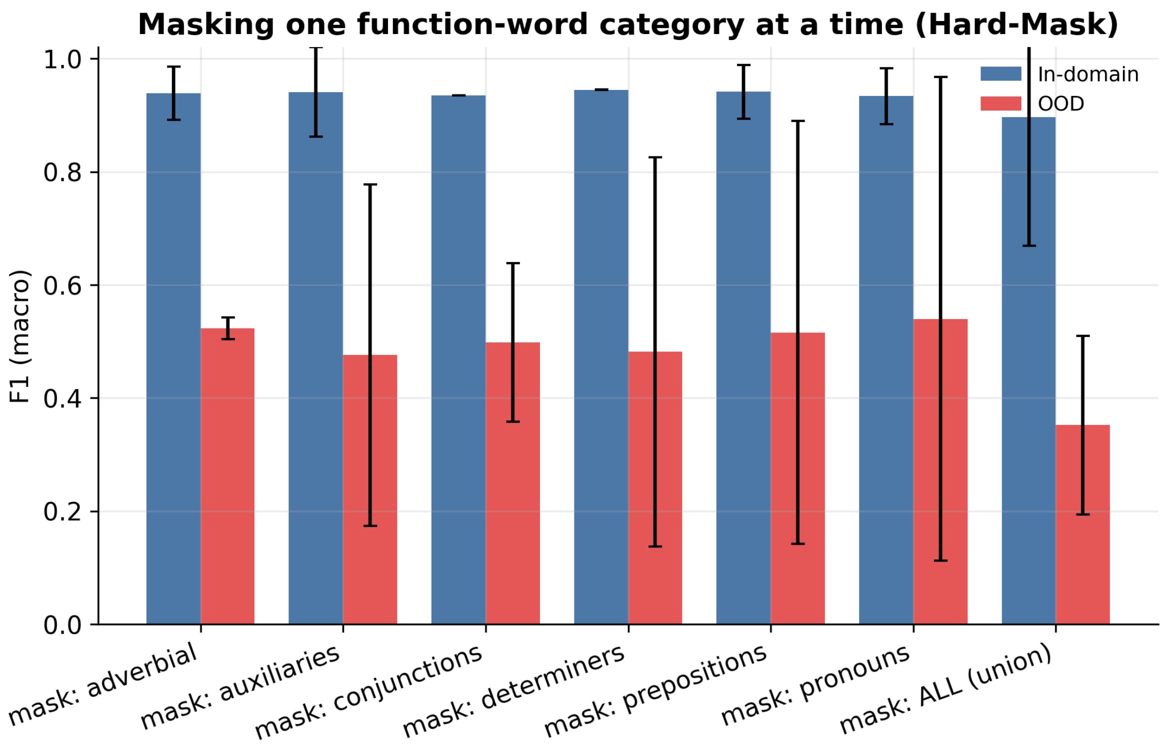

Figure 12: Per-category masking ablation: in-domain vs. OOD F1.

5.8 Few-shot domain recovery

Finally, we ask how much labelled target-domain data is needed to recover the cross-domain loss: the hotel-trained Hard-Mask model is fine-tuned on k labelled out-of-domain reviews and evaluated on a fixed held-out OOD test set.

Table 8: Few-shot domain adaptation of the Hard-Mask model (mean $\pm$ 95% CI over few-shot samples).

Labelled OOD examples k	OOD F1
0	34.3 ± 0.0
50	47.3 ± 14.0
100	58.6 ± 20.9
200	77.2 ± 1.5

Moving from $k = 0$ to $k = 200$ changes OOD F1 from 34% to 77%; the size of this recovery indicates how much of the cross-domain gap is a labelled-data requirement rather than a structural limitation.

5.9 Explanation faithfulness

Table 9: Explanation faithfulness (Integrated Gradients, content words).

Variant	Compr. ↑	Suff. ↓	Del-AUC ↓	Ins-AUC ↑
Baseline	0.100	0.100	0.895	0.891
SoftReg	0.047	0.054	0.900	0.904
Hard-Mask (FAITH)	0.029	-0.000	0.929	0.972

Faithfulness is comparable across variants and mixed in direction (Figure 13): no variant dominates on all four measures, with Hard-Mask strongest on sufficiency and insertion and the baseline stronger on comprehensiveness. Deletion/insertion curves (Figure 14) tell the same story. We therefore do *not* claim a faithfulness improvement from invariance; the explanation contribution is structural—explanations are content-only by construction and computed on the deployed model (Figure 15)—rather than a higher faithfulness score.

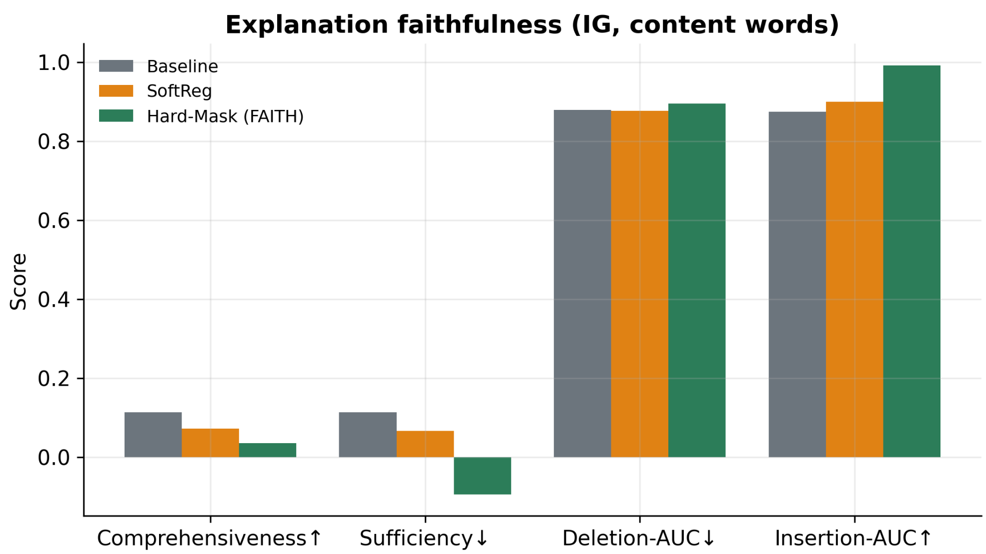

Figure 13: ERASER comprehensiveness/sufficiency and deletion/insertion AUC.

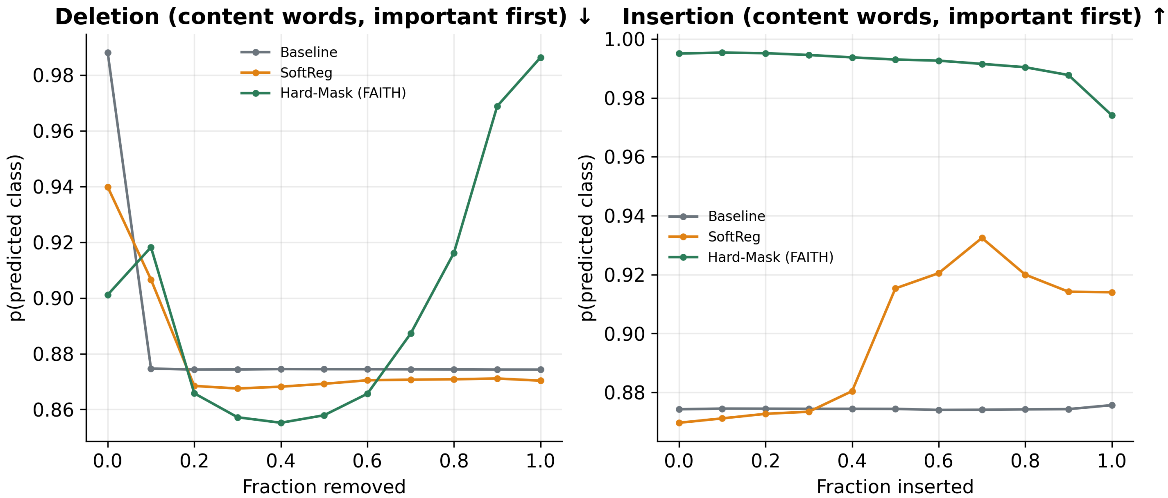

Figure 14: Mean deletion and insertion curves over content words.

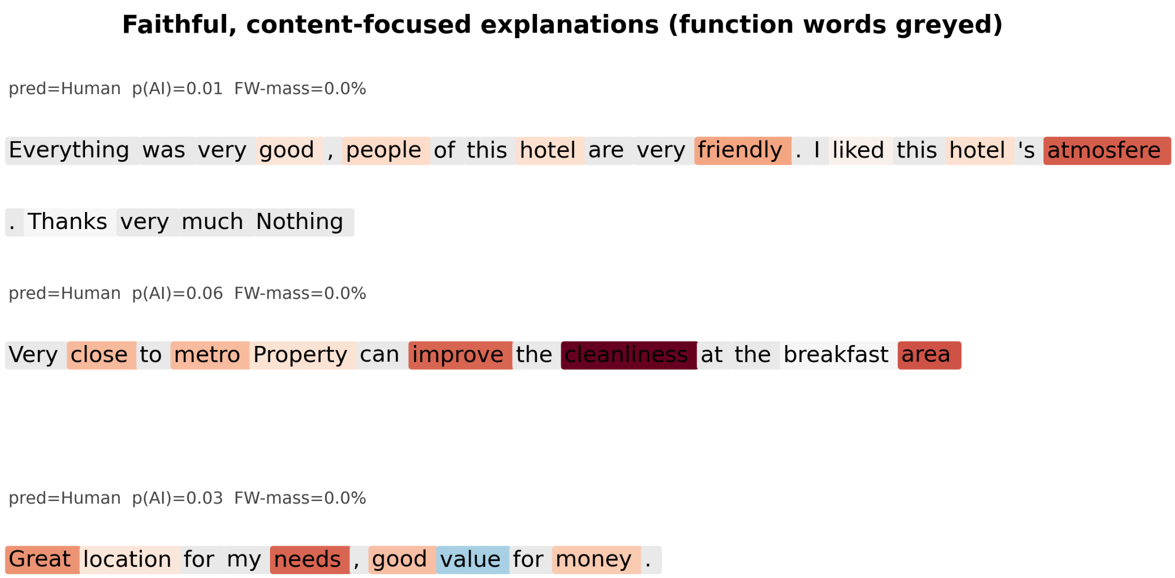

Figure 15: Example explanations: content highlighted, function words greyed.

5.10 Calibration and representation

Reliability diagrams (Figure 16) summarise calibration on the in-domain test set via the expected calibration error [19]; we report it for completeness rather than as a contribution. A two-dimensional projection of the encoder representations (Figure 17) shows that human and AI reviews are well separated in-domain for all variants.

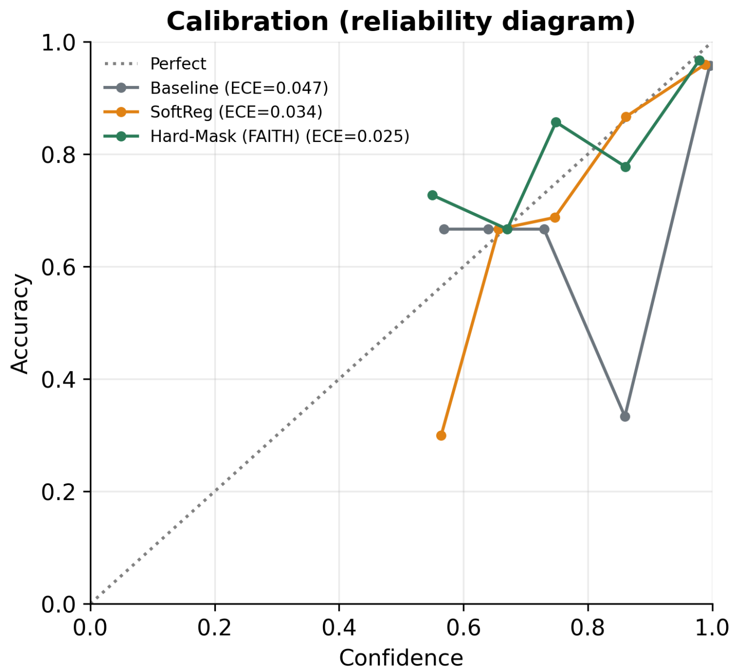

Figure 16: Reliability diagrams (in-domain) with expected calibration error.

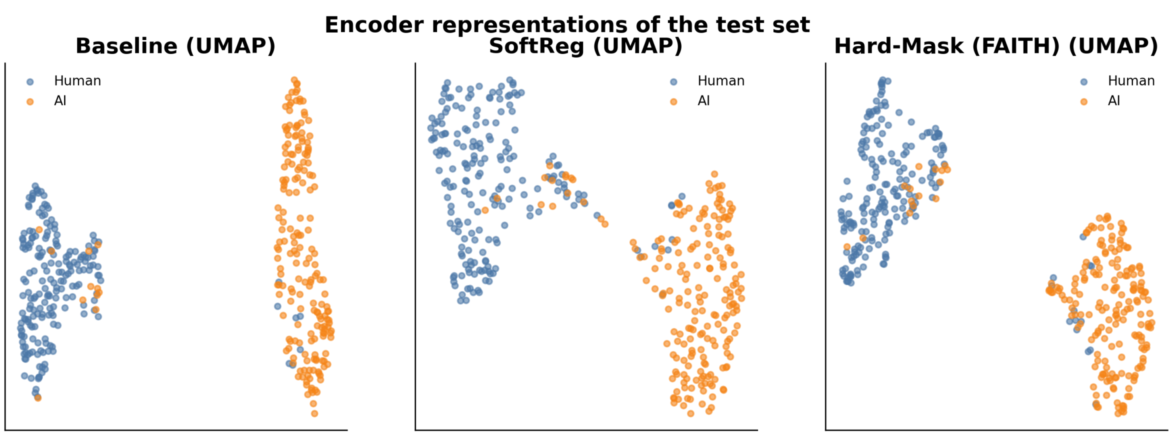

Figure 17: 2-D projection of encoder representations of the test set.

5.11 Leakage

A naive random split inflates baseline F1 to 96.7% versus 93.2% under the grouped, leakage-free split (Figure 18). Because hotels recur with both labels, a random split lets the model exploit hotel-specific content; we report grouped numbers throughout and recommend grouped splits for this dataset.

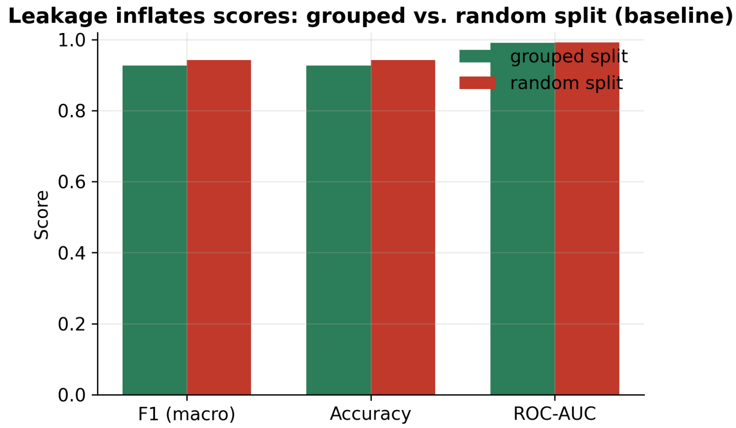

Figure 18: Random vs. grouped split (baseline): random splitting overstates scores.

6 Discussion

Making a detector ignore function words is free in-distribution on this benchmark, yields a provable invariance and the best robustness to the two attacks we test, and produces explanations that are content-only by construction—in contrast to post-hoc, surrogate-based explanations. The cost is cross-domain transfer: when the content vocabulary changes, function-word and stylistic regularities that the constrained model discards carry domain-general signal. Practically, hard masking is attractive where robustness, auditability and a guarantee matter and the domain is fixed; the unconstrained or soft model is preferable for open-domain transfer. We deliberately avoid stronger claims: results are on one in-domain dataset and two attacks, and the OOD evidence is a single benchmark.

7 Limitations

The study is English-only and uses a single in-domain generator (GPT-4), so cross-generator evidence comes from a benchmark that also shifts domain and cannot isolate generator effects. The in-domain set is modest (2,000 reviews), which widens confidence intervals; we report them honestly rather than selecting favourable seeds. The two attacks are simple proxies for adversarial and paraphrase pressure, not an exhaustive robustness audit. Faithfulness metrics are themselves contested [14] and we do not treat any single one as decisive. Function-word invariance is a design choice with a measured trade-off, not a universally preferable configuration.

8 Conclusion

Function words are a measurable shortcut in AI-text detection. A detector that provably ignores them keeps in-domain accuracy, gains robustness to the attacks we test, and explains itself with content alone—at a measured cost to cross-domain transfer. We release code, leakage-free protocols, multi-seed results with confidence intervals, and figures that regenerate from a single results file.

Reproducibility. All numbers come from a single results JSON; figures and this paper regenerate from it; the hard-masking guarantee is asserted by an automated test. Code: `https://github.com/scar09-22/FAITH-Detect`.

A Training behaviour

Figure 19 shows training loss and validation F1 by epoch, averaged over seeds. The soft and hard variants start slower (they discard or down-weight function-word information) but reach comparable validation F1.

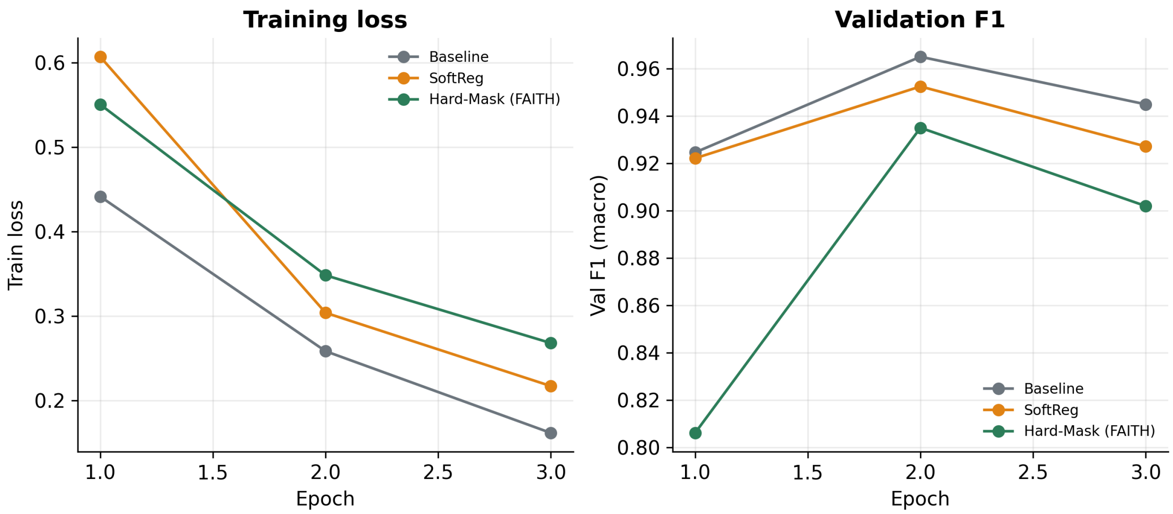

Figure 19: Training loss and validation F1 by epoch (mean over seeds).

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